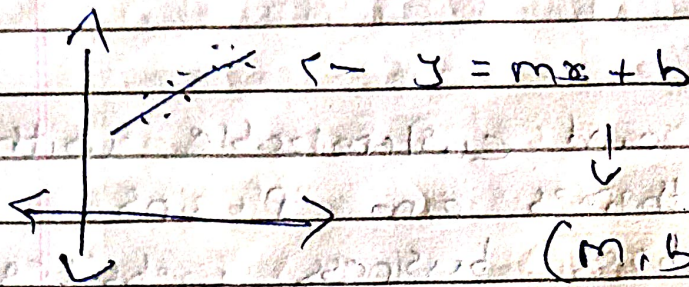


Maths behind

Linear regression



Closed form

Solution

OLS

(Ordinary least squares)

Non-closed form

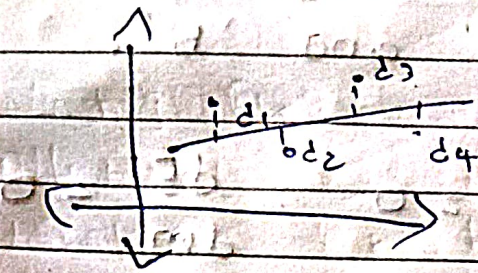
Solution

Gradient descent

* Formula

$$b = \bar{y} - m\bar{x}$$
$$M = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$
$$\sum_{i=1}^n (x_i - \bar{x})^2$$

* Maths behind Formula



$$E = d_1 + d_2 + d_3 + \dots + d_n$$

$$E = d_1^2 + d_2^2 + \dots + d_n^2$$

$$E = \sum_{i=1}^n d_i^2$$

Error
function
(or)

Loss
function

$$\therefore d_i = (y_i - \hat{y}_i)^2$$

$$E = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\hat{y}_i = mx_i + b$$

$$E(m, b) = \sum_{i=1}^n (y_i - mx_i - b)^2$$



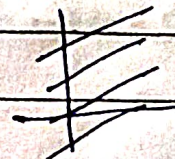
Now, ...

$$b = 0$$



$$1) E(m) = \sum_{i=1}^n (y_i - mx_i)^2 \dots \text{--- (1)}$$

$$m = 0$$



$$2) E(b) = \sum_{i=1}^n (y_i - x_i - b)^2 \dots \text{--- (2)}$$

→ we will find derivative of Error function in respect to 'm' and 'b'.

$$F(m, b) \quad \frac{dF}{db} = 0, \quad \frac{dF}{dm} = 0$$

1. b

$$\frac{dF}{db} = \frac{d}{db} \sum_{i=1}^n (y_i - mx_i - b)^2 = 0$$

$$= \sum_{i=1}^n \frac{d}{db} (y_i - mx_i - b)^2 = 0$$

$$= \sum_{i=1}^n 2(y_i - mx_i - b) = 0$$

$$\sum (y_i - mx_i - b) = 0$$

$$\sum \underline{y_i} - \sum \underline{mx_i} - \sum \underline{b} = 0$$

$$\sum y_i - m \sum x_i - nb = 0 \Rightarrow \sum y_i - m \sum x_i = nb$$

$$\sum y_i - m \sum x_i = nb$$

$$\boxed{b = \bar{y} - m\bar{x}}$$

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2. 3

$$E = \sum (y_i - mx_i - \bar{y} + m\bar{x})^2$$

$$\frac{dE}{dm} = \sum_{i=1}^n \frac{d}{dm} (y_i - mx_i - \bar{y} + m\bar{x})^2 = 0$$

$$= \sum_{i=1}^n 2(y_i - mx_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) = 0$$

$$= \sum_{i=1}^n -2(y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x}) = 0$$

$$= \sum_{i=1}^n (y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x}) = 0$$

$$= \sum_{i=1}^n ((y_i - \bar{y}) - m(x_i - \bar{x}))(x_i - \bar{x}) = 0$$

$$= \sum_{i=1}^n [(y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2] = 0$$

$$\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) = m \sum_{i=1}^n (x_i - \bar{x})^2$$

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$